Project Review: Remaining useful life Modeling

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1 Introduction

In industry 4.0, Downtime of the production machine causes massive disruptions to the production and cause heavy losses such as cost of maintenance, production stoppage and many more. It is important to address this problem and artificial intelligence has given an acceptable solution to solve this problem. This is done by applying the AI models to learn the behavior of machines that have gone defect and machine that are working properly.

2 Dataset Description

The dataset is a telemetry sensor data of 96 machines where the description of each machine has not been provided. Therefore, the assumption has been made is that all the machines or most of the machines are unique and different and produces a unique sensor data distribution. The assumption is very import for the 2 model configurations. The data set contains 4 types of sensor data for all 96 machines and they are the following:

1. Temperature
2. Pressure
3. Vibration
4. Humidity

3 Data Preprocessing

The dataset contains a very important column called **MachineID** where it acts like a unique identifier for each machine. This variable has allowed the modelling to be in 2 different model configurations and they are the following:

1. **Factory specific model configuration:** In this model configuration, The assumption is that the trained model is going to be deployed in a factory to assess the existing machine where the model has already seen the historical data of each machine and is ready to predict and monitor those models in a factory. The idea is that the model will constantly receive the 4 sensor data and then upon receiving those sensor values the model will assess them and predict a remaining time for the machine to go under maintenance.

2. **Generalized model configuration:** In this model configuration, The assumption is that the model is being trained as a commercial product where the model has been trained without the MachineID variable in a fully generalized manner. With the assumption that all machines are different or most of them are different, The model will be deployed for multiple commercially sold machines.

4 Exploratory Data Analysis (EDA)

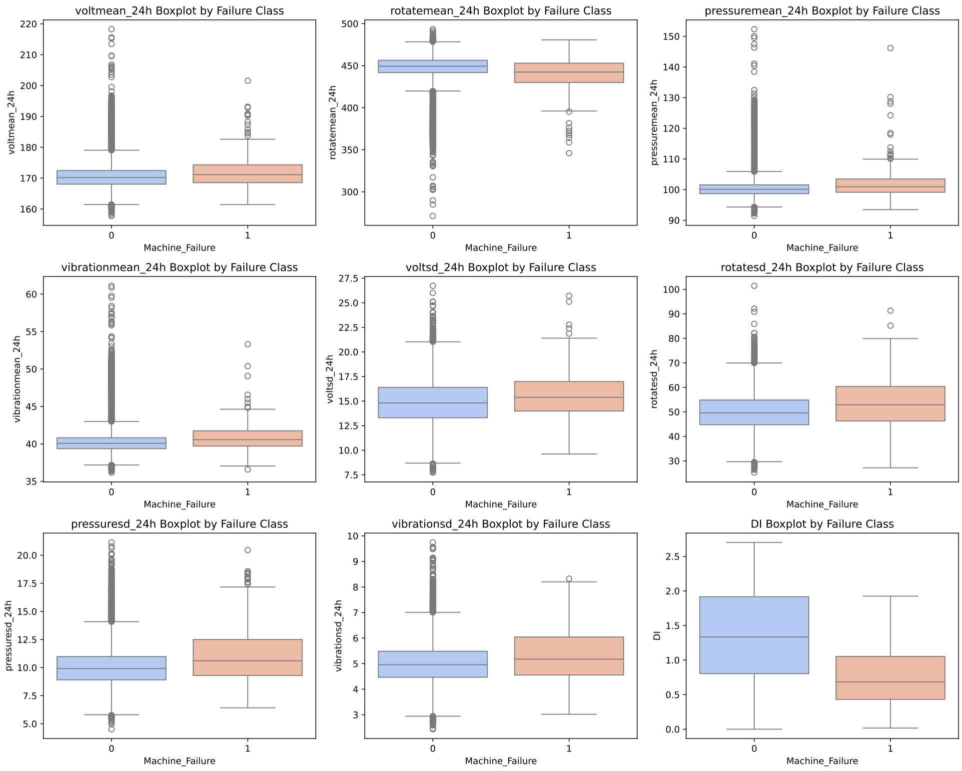


Figure 1: My plot

In this above grid box plots, There are 2 boxes in each plot where one is the sensor reading distributional box plot of healthy working machine values and the other orange box plot is the sensor values box plot where the machine has gone faulty. In all sensor value boxplots, The boxes lie almost has the positional boxes which means that sensor values for both scenario has very little discriminative power for both classes but in the DI(Degradation Index) plot which is a single statistic or point value achieved from all 4 sensor values represents all 4 sensor values, resulting in well separation of both classes.

This phenomenon can also be seen in the SHAP explanation further in the report, where DI has one of the highest mean SHAP values representing that the combined statistic of all 4 sensors contributes very highly towards the prediction.

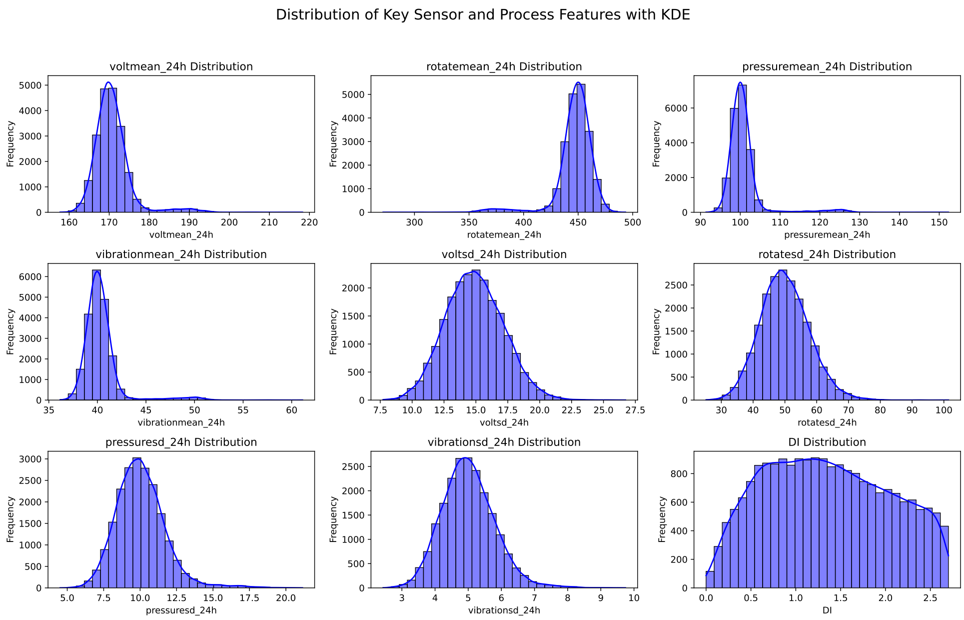


Figure 2: My plot

The above grid plot represents the distributions of predictor sensor variables where each sensor has a specific distribution. There are 2 important observations from these distribution plots.

1. In all sensor distribution plots, 3 out of 4 sensor plots has 2 curves in the plots, One is the bigger bell shaped curve, and the other is very minimally visible where there is a small curve further in the plot for all 3 sensors. My assumption is that this small curve represents the faulty machine sensor values which are hard to interpret form the previous box plots.

2. In the Degradation Index plot, the distribution is almost identical to an uniform distribution plot where the both ends of the distributions has lower frequency of values representing the machine faults and almost consistent length of distribution in the middle representing the heathy performance of the machines

5 Feature Engineering

There are 2 different data modalities available in the dataset. One is the sensor values taken for every 24 hour format, and another is the aggregated score of 5 day sensor value format. The models have been fit for both of these data separately for response variable.

6 Modeling

In this project, The availability of the MachineID variable has allowed for 2 different model configurations where there has been an advanced analysis for each model configuration has been done. The both model configurations are the following:

1. Factory-specific modelling.

2. Generalized modelling.

6.1 Models Used

For **Factory-specific modelling**, As mahcineID represents a unique identifier for the machines, It needs to be efficiently modeled. One such model where the inclusion of the machineID variable as categorical variable which is **CatBoost**. It must be made sure that if the unique identifier is a numerical variable then it must not be given as a predictor variable to any regression models because the model would consider that numerical identifier as a normal rpedictor variable and cannot properly understand the specific behaviour nof each machine when the machineID is a integer for each machine and the analysis from such a model would not be reliable. This is where CatBoost solves this problem by having a hyperparameter that takes the categoirical variable and efficiently models it.

For **Generalized modelling**, As the machineID variable has not been included in this modelling, the generalization becomes much difficult as the assumption is that all or almost all machines are different, they produce different distributions of sensor values and to generalize all those distributions becomes much harder. As there is a high non linear relations, The idea is to see the fitting of non parametric and non linear models such as the following:

1. Decision Trees
2. Random Forests
3. CatBoost
4. XGBoost
5. Multilayer Perceptron

6.2 Evaluation Metrics

The evaluation metrics that are selected for both configurations are the following:

1. **MAE:** The mean absolute error is an easy to interpret metrics that can give a single valued statistic summarizing whole model performance.
2. **RMSE:** The root mean squared error metric provides an in depth model performance analysis where it an idea not just about the whole model performance but also about the accuracy of high error prediction. It is specifically useful in the remaining useful modelling where the difference between prediction and actual values for all cycles is desired and RMSE summairzes that.
3. $R2$ **score:** The R2 score is the proportion of target variable variance explained by the model. It acts like a model accuracy where high values represents perfect model predictions.

7 Results

There are 7 different classifiers are used to fit to the data where the metrics mae,mse,R2 are used for further model selection. Out of all the classifiers, Catboost regression model has given the best results with respect to the etrics where the r2 score for this model was 0.86 which mean the model is able to explain the 86% percent of the variance form the data. Other metrics such as the mean absolute score and mean squared error both were low with the values 6 and 20 respectively. The reason for this level of model performance is the sufficient data availability where the model is able to learn meaningful insights from the large amount of data andalso the internal mechanism of the catboost model itself allows it to learn difficult non linear effects on the target variable. But the previously discussed metrics are no enough for model selection where there is always a possiblity that the model come across the data whcih it has not seen before in terms of the different distrbution of data due to irregularities in the data collection, or data corruption where the acutual data changes a bit. The mdoel should be robust enough to withstand these differences and predict the produce the pre production performance all the time and this can be analysed witht he robustness testing hwere the testing data gets injected with gaussian noise of selected percentage and then the model is predicted for this new data and comparison witht he normal preidction would reveal the picture of robustness. After this test, the catboost model for 5 percent gaussian noise injected testing data has leanred 75% variance which is a satisfactory results given the data is purposefully corrupted and still it is able to give reliable performance.

7.1 $r2$ score accuracies of both model configurations:

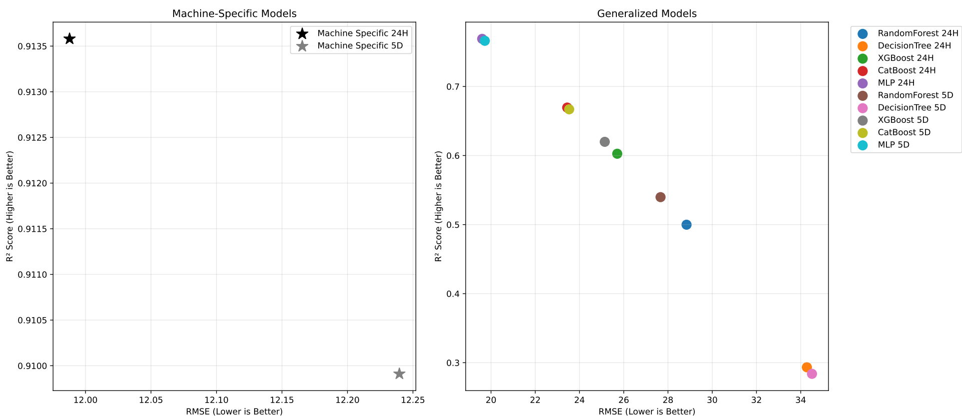


Figure 3: My plot

The above plot represents the $r2$ scores of all the models in both model configurations. The left plot is for the machine known model configuration and the right is for machine unknown model configuration. The x axis is the RMSE score and y axis is the $r2$ score. In machine known modeling, a catboost model has been used to fit the data, and as there are 2 different data types, both RMSE, and $r2$ scores have been plotted for the catboost models trained on both data types separately. Both models have shown the best performance and the different is very minimal, but in both the RMSE, and $r2$ score, the clear winner with a very slight margin is the catboost model trained on the 24 hour sensor data with 0.9125 $r2$ score and 12 days RMSE score,delivering almost perfect result. The reasoning behind this could possibly be the details in the sensor values of each day could be the deterministic space where the model learns those important underlying classification features accurately.

The right plot is also the same plot but in a generalized setting where the machine ID columns has not been included while modelling and has been used to predict in a generalized manner. In this setting, 5 models have been used and 2 data types. So for analysis, a total of 10 models have been trained and analyzed. Out of all, Multilayer perceptron with 24hour datatype has given the best results with 0.74 $r2$ score and 20 days RMSE. As these models have been trained in a generalized way, the metrics are lower compared to the models from the machine known setting. The assumption is that not all 100 machines are of same type which would results in different kinds of sensor values, and the models are having trouble in finding that classification line for all 100 machines efficiently. But multilayer perceptron has shown very promise in this type of setting due to its powerful nature of capturing non linear behavior.

7.2 Machine known modelling:

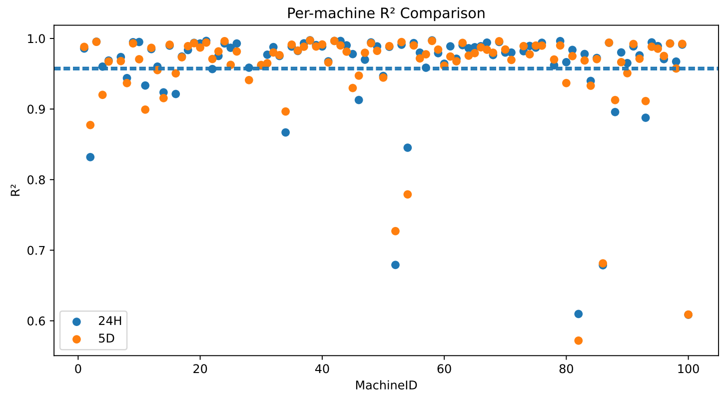


Figure 4: My plot

In this, the $r2$ score of all 100 machine from the machine known modeling has been plotted. It is to make sure that the high $r2$ score is not dominated by some outliers. As the in the plot, almost 90 percent of machines have an $r2$ score of over 0.95 showing that the model performance is consistent across all machines.

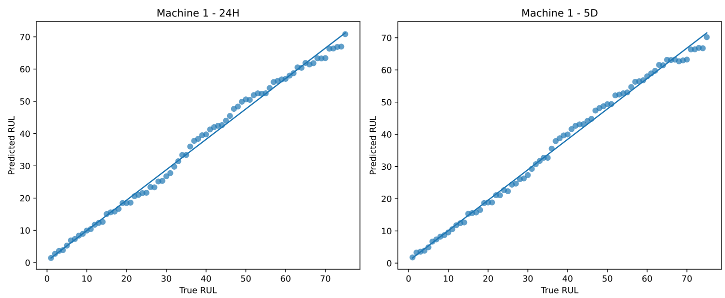


Figure 5: My plot

The above plot is the scatter plot of True RUL and predicted RUL for the 1st machine and the scattered points follows the line accurately. The model stays consistent from low RUL prediction to high RUL prediction.

7.3 Machine unknown model configuration:

In this sub section, the machine unknown model results are discussed. For this modeling, 5 models have been fit and analyzed using visual plot analysis. The following are the scatter plot between true RUL and predicted RUL of 5 models across 2 different data types. Decision tree and random forest models are shown to be the weakest models across all. The scatter plots reveal that the under predictions form both models and the behaviors of the scattered points also deviated heavily form the perfect line. One reason could be is that the models are too simple for the generalized setting. As there could be different types of machines, the tree based models are too simple for generalization and delivering under predictions. It is a completely different case for the other 3 models where all 3 are delivering good performance and the scattered points are staying close to the perfect line, depicting good generalization.

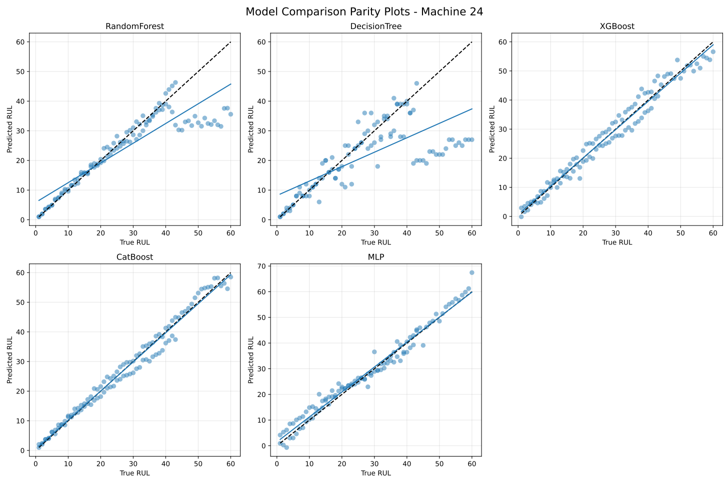


Figure 6: My plot

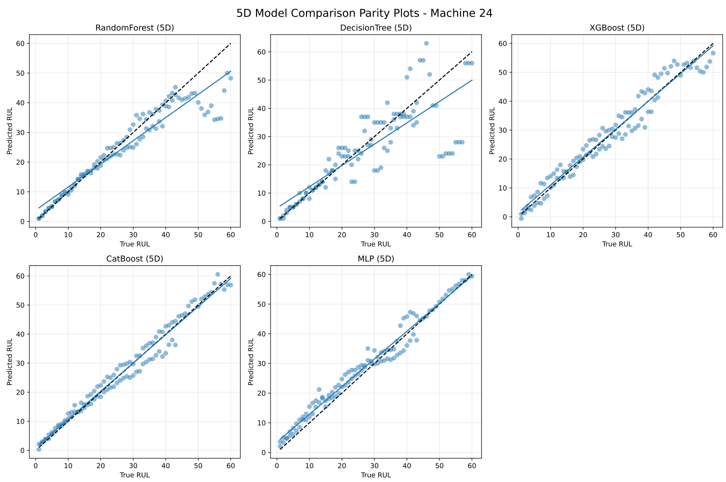


Figure 7: My plot

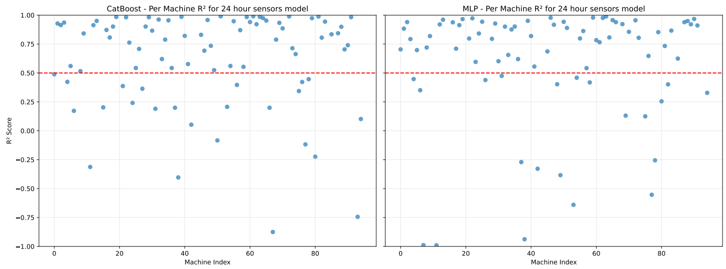


Figure 8: My plot

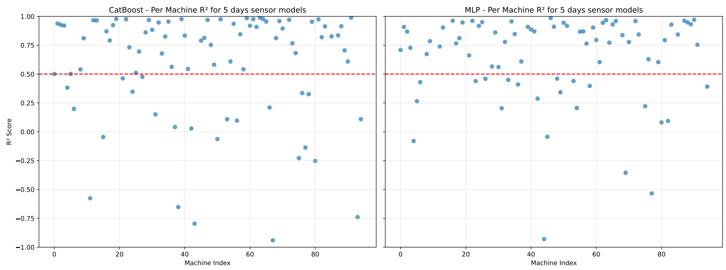
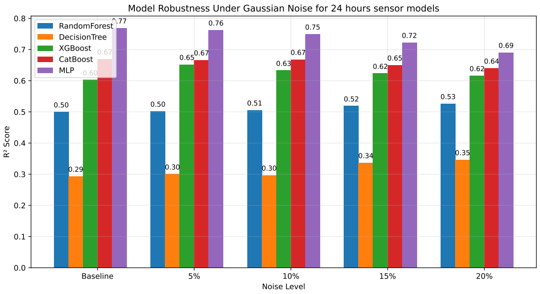


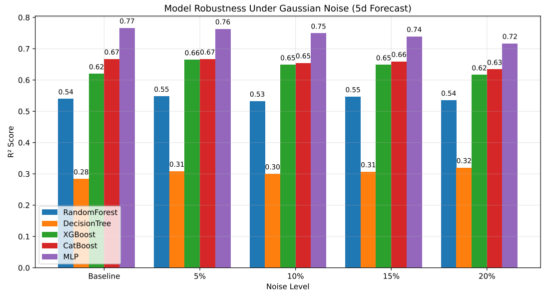
Figure 9: My plot

The following plots are the $r2$ scores per machine in the machine unknown model configuration. The left plot is form the catboost model and the right plot is of multilayer perceptron model. Compared to catboost and also other 3 models, Multilayer perceptron has delivered more than 0.5 $r2$ score for 80% of the machines which is significant given the generalized model conifguration. One importantstudy that can be done is to see the machines that achieved the lower $r2$ scores are same in both machine known and unknown settings. If the machines are same then these machine would be acorss modelling setting, have difficulty in generalizing. hi so this is me writing first time on my ipad using a bluetooth keyboard and mouse which is awesome and amazing at the same time because now i can work the general work on my ipad using these 2. which makes it very efficient and amazing

7.4 Robustness testing:



(a) First plot

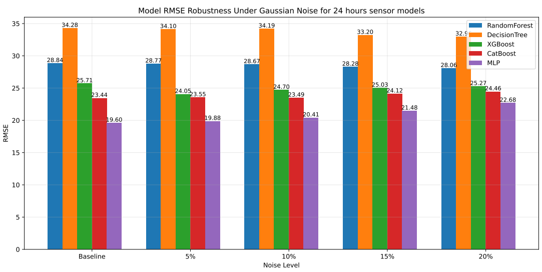


(b) Second plot

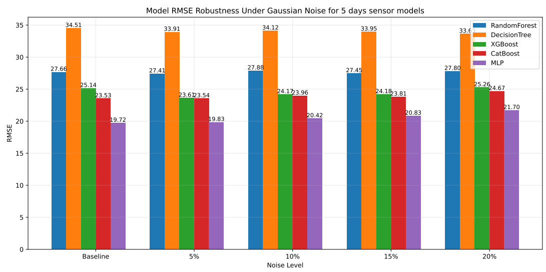
Figure 10: Two plots side by side

Robustness testing has been done to see the model performance degradation with the data being injected with noise. The assumption is that the sensor data is not accurate and may often contain real world noise and models learning meaningful rules form these noisy data and how the predictions are from these models is the goal to analyses in this section. It also analyses the question, how the models are withstanding the increasing level of noisy data. If the drop is higher and higher as the noise levels increase then the model is unstable and cannot truly generalize for real world unseen data.

In the above histogram plot, 2 models has shown significantly consistent performance where the degradation stayed relatively very low. They are Catboost and MLP. One interesting behavior is from the decision tree and random forest where instead of the $r2$ score degrading , it got better than the baseline. But is is not surprising as the reason for this behavior could is that the added noise might be acting as a regularizing tool for better generalization for these 2 models which ca also be confirmed from the following RMSE plots.



(a) First plot



(b) Second plot

Figure 11: Two plots side by side

RMSE metric is an important statistic which heavily penalizes the error predictions and is a very useful metric to know more than just average prediction error. Among all the models, It is MLP which has the lowest RMSE score but it is important address that it is not the best stable model when trained and tested with noisy data. Although Decision tree and random forest has the highest RMSE scores, They are being lowered with increase in the noise level, showing similar behavior from the $r2$ score robustness analysis plot above. Other models such as catboost and XGBoost showed extremely stead results and It is only MLP which still has the highest RMSE even with 20 percent noise level compared to all the other baselines, It has increased the RMSE steadily. The increase is not significant as a stand alone model but is considerable compared to other fitted models.

The following are the smooth plots of $r2$ scores per observation. These provide clear depiction of the stability in a clear way, where in the both plots, MLP model has shown the matching of baseline, 5% noise and 10% noise models almost exactly same across observations. the significant drop is only at 15% and 20% which is expected given the high noise levels

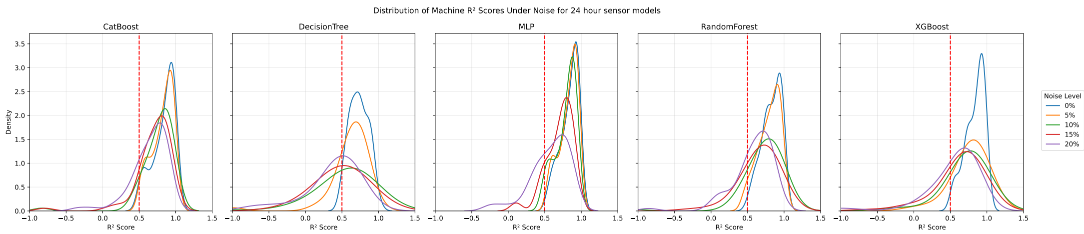


Figure 12: My plot

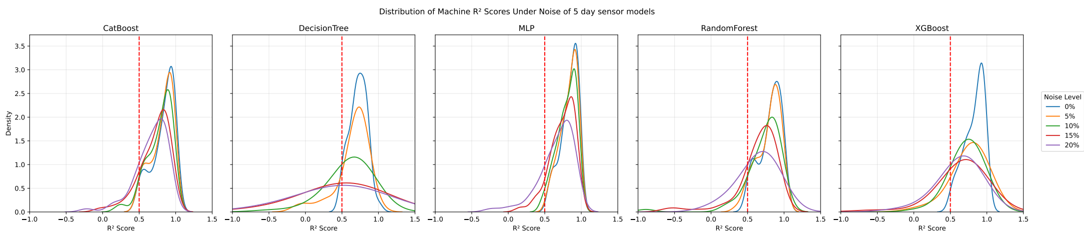


Figure 13: My plot

7.5 Explainable AI:

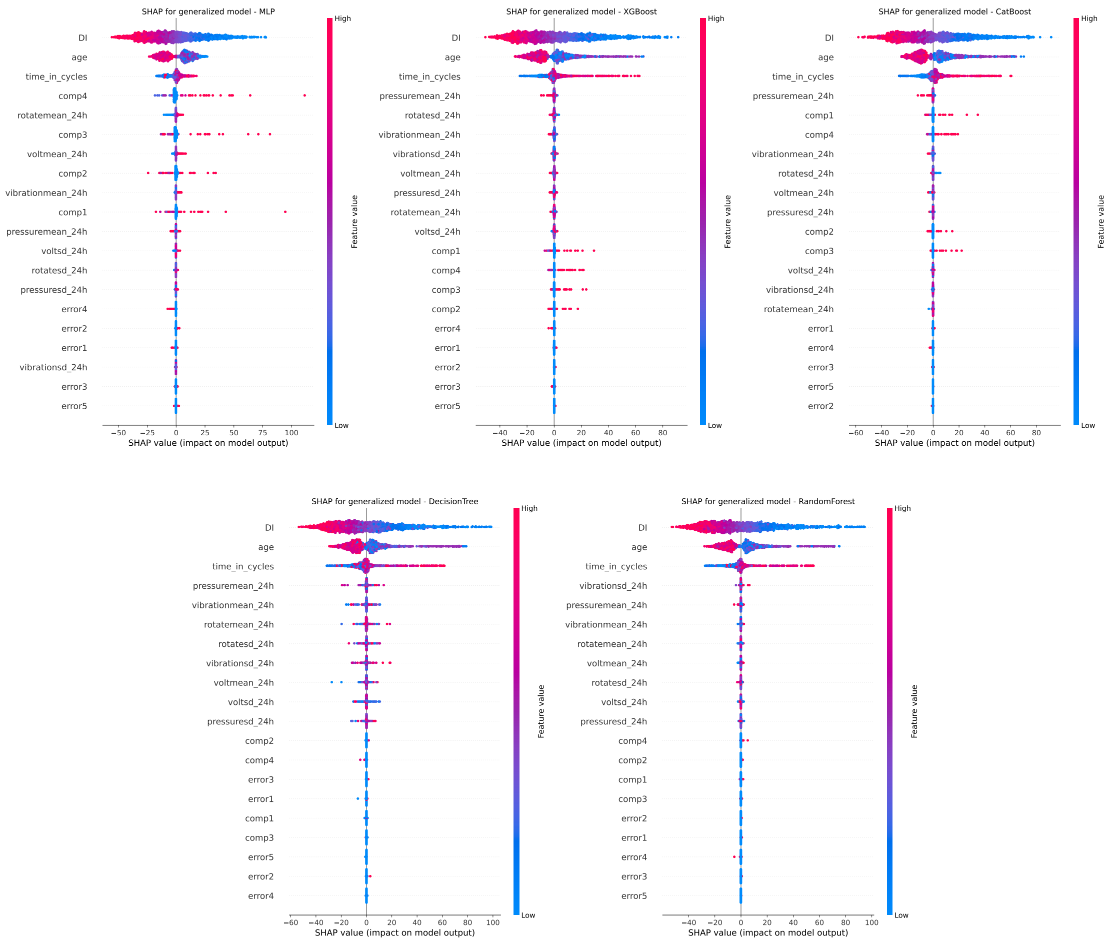


Figure 14: Five plots arranged in a grid

In the above grid of plots, all 5 plots are the beeswarm SHAP explanation plots. These are for the machine unknown model configuration plots. One constant behavior in all 5 plots is that the top 3 variables remain same. The series remains the same and the only difference is in the magnitude. The top variable is the degradation index(DI) which is defined from all the the sensor values. It acts like a single metric comprising the summarized data of all sensors. Second is the age of the machine itself and the final one is the time in cycles representing the functioning cycle of the machine. The variables after three are highly flexible for each model. Each model has it's own compabination of variables from 4th position. The model can get significant amout of udnerlying features just from these 3 variables, promising a significant future evolution of efficient handling of model size and comparability. [Latent dimensionality] Consistently important features. meaning that these 3 features hold the the latent features that can be used for classification alone.